Customer Churn Prediction for a Telecom Company Using Machine Learning

Internship Project Report

Project Type: Individual Project

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Abstract

Customer churn, the decision of a customer to stop using a company's services, directly affects the revenue and growth of telecom companies. This project analyses a dataset of 7,043 telecom customers to identify the patterns associated with churn and to build machine learning models that estimate the likelihood of a customer leaving. The work covers data quality assessment, exploratory data analysis, statistical validation, feature engineering, model comparison, model evaluation and interpretation of results.

Two classification models, Logistic Regression and Random Forest, were trained and compared against a majority-class baseline. Logistic Regression achieved the best F1-score (0.613) and ROC-AUC (0.842), and also the highest churn recall (0.794). Random Forest achieved the highest accuracy (0.764) and precision (0.547). Contract type, tenure, charges and payment method were found to be strongly associated with churn. The trained model was saved and a ranked list of customer churn probabilities was exported to Excel so that retention teams can act on it.

1. Introduction

1.1 Background

Customer churn refers to a situation in which a customer decides to stop using the services of a company. It is an important problem for telecom companies because losing existing customers reduces revenue and slows growth. Identifying at-risk customers early allows a company to respond with support or retention offers before the customer leaves.

1.2 Project Objective

The objective of this project is to analyse customer characteristics, identify patterns associated with churn, and develop a machine learning model that estimates the likelihood of a customer leaving the service.

1.3 Business Value

The analysis helps telecom companies to:

- identify customers who may need additional support;
- investigate the possible reasons for churn;
- design targeted customer retention strategies.

1.4 Analytical Approach

The project follows a structured workflow: data quality assessment, exploratory data analysis, statistical validation, feature engineering, model comparison, model evaluation and interpretation of results.

1.5 Scope of the Dataset

The project uses a historical snapshot of customer information. The dataset supports customer-level churn classification, but it does not provide a complete sequence of dated customer events, so it cannot be used to forecast churn over time.

2. Dataset Description

The Telco Customer Churn dataset (published by IBM) was loaded directly from its public GitHub source. It contains **7,043 customers** and **21 columns**: a customer identifier, 19 input features and the target variable.

2.1 Input Features

Category	Features
Demographics	Gender, Senior Citizen, Partner, Dependents
Phone and internet services	Phone Service, Multiple Lines, Internet Service
Add-on services	Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies
Account and billing	Contract, Paperless Billing, Payment Method
Numeric account values	Tenure, Monthly Charges, Total Charges

2.2 Target Variable

The target variable is **Churn**. It is encoded as 0 (the customer will stay) and 1 (the customer will churn).

3. Tools and Technologies

Tool	Purpose
Python	Programming language
Jupyter Notebook	Development environment
pandas, NumPy	Data loading, cleaning and manipulation
Matplotlib, Seaborn	Data visualisation
SciPy	Chi-square statistical test
scikit-learn	Preprocessing pipelines, models, cross-validation and metrics
openpyxl	Exporting predictions to Excel
joblib	Saving and loading the trained model

4. Methodology

The project was carried out in the following stages:

1 Import the required libraries and load the dataset.
2 Understand the dataset and check its quality.
3 Clean the data and engineer new features.
4 Explore churn patterns using visualisations.
5 Validate the findings with a statistical test.
6 Separate the features and target, then split the data into training and test sets.
7 Build a preprocessing pipeline and train the models.
8 Compare the models with a baseline, evaluate them and validate them using cross-validation.
9 Interpret the important features and generate predictions.
10 Save the model and export the results.

5. Data Understanding and Cleaning

5.1 Data Quality Checks

The dataset was checked for structure and integrity before any modelling was done.

Check	Result
Shape	7,043 rows and 21 columns
Duplicate rows	0
Duplicate customer IDs	0
Negative tenure, monthly charges or total charges	0
Missing values (initial)	0

5.2 Data Type Correction

The TotalCharges column was stored as text. It was converted to a numeric type, and values that could not be converted became missing values. After conversion, **11 records** had missing TotalCharges. These were handled inside the modelling pipeline using median imputation for numeric columns.

6. Feature Engineering

Two new features were created to describe customer behaviour more richly:

- **AverageChargePerTenureMonth**: Total Charges divided by Tenure, giving the average amount a customer has paid per month of service. Records with zero tenure were treated as missing to avoid division by zero.
- **HasMultipleServices**: a count of how many of seven services a customer subscribes to (phone service, online security, online backup, device protection, tech support, streaming TV and streaming movies).

A third feature, **TenureGroup**, was also derived by grouping tenure into four bands: 0–12, 13–24, 25–48 and 49–72 months. After feature engineering the model used 22 input features: 6 numeric and 16 categorical.

7. Exploratory Data Analysis

7.1 Churn Distribution

The target classes are imbalanced. Of the 7,043 customers, 5,174 stayed and 1,869 churned.

Class	Customers	Share
Stayed (No)	5,174	73.46%
Churned (Yes)	1,869	26.54%

Because of this imbalance, accuracy alone is a misleading measure of quality, and recall, precision, F1-score and ROC-AUC are reported alongside it.

7.2 Churn by Customer Tenure

Churn is highest among new customers and falls steadily as customers stay longer.

Tenure group	Churn rate
0–12 months	47.44%
13–24 months	28.71%
25–48 months	20.39%
49–72 months	9.51%

Almost half of customers in their first year left, compared with under 10% of customers who had stayed more than four years. The first year is therefore the most critical period for retention.

7.3 Churn by Payment Method

Payment method	Churn rate
Electronic check	45.29%
Mailed check	19.11%
Bank transfer (automatic)	16.71%
Credit card (automatic)	15.24%

Customers paying by electronic check churn at roughly three times the rate of customers using automatic payment methods.

7.4 Churn by Contract Type

A count plot of churn by contract type (month-to-month, one year and two year) was produced in the notebook. The statistical test in the next section confirms that contract type and churn are strongly related, and the model also ranks month-to-month contracts among its most important features.

8. Statistical Validation

A chi-square test of independence was applied to check whether contract type and churn are related.

Measure	Value
Chi-square statistic	1184.597
Degrees of freedom	2
P-value	5.86×10^{-258}

The p-value is far below 0.05, so there is a statistically significant association between contract type and churn. This is evidence of association only, and it does not prove that contract type causes churn.

9. Model Development

9.1 Train–Test Split

The data was split 80% for training and 20% for testing using a stratified split (random state 42), so that the churn proportion is the same in both sets.

Set	Records
Training	5,634
Testing	1,409

9.2 Preprocessing Pipeline

Preprocessing was built as a scikit-learn pipeline so that exactly the same steps are applied during training, testing and prediction, which prevents data leakage.

- **Numeric features**: median imputation followed by standard scaling.
- **Categorical features**: most-frequent imputation followed by one-hot encoding, with unknown categories ignored.

9.3 Models Trained

Model	Configuration
Baseline (Dummy Classifier)	Always predicts the majority class (customer stays)
Random Forest	200 trees, balanced class weights, random state 42
Logistic Regression	Balanced class weights, up to 5,000 iterations

Balanced class weights were used in both models so that the minority class (churners) is not ignored during training.

10. Results and Evaluation

10.1 Baseline Performance

The baseline model reached an accuracy of 0.735 but a churn recall of 0.000, since it never identifies a single churner. This shows why accuracy alone is not a sufficient measure for this problem and sets the minimum standard the real models must beat.

10.2 Model Comparison on the Test Set

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.734	0.499	0.794	0.613	0.842
Random Forest	0.764	0.547	0.636	0.588	0.822

Logistic Regression is the better model for finding churners: it has the highest recall, F1-score and ROC-AUC. Random Forest is more accurate overall and more precise, meaning fewer of the customers it flags are false alarms. The better choice depends on the business cost of missing a churner versus contacting a customer who would have stayed.

10.3 Random Forest Classification Report

Class	Precision	Recall	F1-score	Support
Stayed	0.86	0.81	0.83	1,035
Churned	0.55	0.64	0.59	374
Overall accuracy			0.76	1,409

The Random Forest identifies about 64% of customers who actually churned, with roughly 55% of its churn predictions being correct. A confusion matrix for this model was plotted in the notebook.

10.4 Cross-Validation

The Random Forest was validated with 5-fold stratified cross-validation on the training data. Results are reported as mean ± standard deviation.

Metric	Mean ± Std. dev.
Accuracy	0.779 ± 0.013
Precision	0.574 ± 0.022
Recall	0.643 ± 0.034
F1-score	0.607 ± 0.027
ROC-AUC	0.829 ± 0.010

The low standard deviations show that the model performs consistently across different subsets of the data, and the cross-validated scores are close to the test-set scores, which suggests the model is not strongly overfitted.

11. Model Interpretation

Feature importance from the Random Forest was used to see which customer attributes contribute most to the predictions.

Rank	Feature	Importance
1	Total Charges	0.110
2	Tenure	0.100
3	Average Charge per Tenure Month	0.098
4	Monthly Charges	0.096
5	Contract: Month-to-month	0.077
6	Online Security: No	0.033
7	Tech Support: No	0.029
8	Internet Service: Fiber optic	0.027
9	Contract: Two year	0.026
10	Payment Method: Electronic check	0.024

The most influential features are the financial and tenure measures, followed by contract type, the absence of security and support services, fibre-optic internet and electronic-check payment. These results agree with the exploratory and statistical findings. Feature importance measures predictive contribution, not causation, so these results should be interpreted cautiously and only within this dataset.

12. Predictions and Deliverables

12.1 Individual Customer Prediction

The trained model can score any individual customer. For a sample customer from the test set, it predicted that the customer would stay, with an estimated churn probability of 0%.

12.2 Excel Export

Predictions for all 1,409 test customers were exported to an Excel file, `telecom_churn_predictions.xlsx`, containing the customer ID, actual churn, predicted churn and churn probability (%), sorted from highest to lowest risk. The highest-risk customers had probabilities of 98.5% to 100%, and most of the top ten were customers who actually churned.

12.3 Saved Model

The trained Random Forest pipeline was saved as `telecom_churn_model.pkl` using joblib and reloaded successfully, so it can be reused without retraining.

12.4 Project Summary

Item	Value
Total customers	7,043
Input features	22
Training records	5,634
Testing records	1,409
Best model by F1-score	Logistic Regression
F1-score	0.613
ROC-AUC	0.842

13. Business Recommendations

Based on the patterns found in this analysis, the following actions could be tested as retention experiments:

- **Focus on the first year.** Nearly half of customers in their first 12 months churned, so onboarding support and early check-ins are likely to have the greatest effect.
- **Encourage longer contracts.** Month-to-month contracts are strongly associated with churn, so incentives to move customers to one- or two-year plans may help.
- **Promote automatic payments.** Electronic-check customers churn far more than customers on automatic payment methods.
- **Promote security and support add-ons.** Customers without online security or tech support are more likely to churn.

- **Review fibre-optic customers.** Fibre-optic internet appears among the most important features and deserves a closer look at pricing and service quality.
- **Use the risk list.** The ranked churn-probability list lets retention teams contact the highest-risk customers first.

These are suggestions drawn from associations in the data. They should be tested before being treated as proven causes.

14. Limitations

1 The analysis uses a public sample dataset and may not represent every telecom company's customer base.

2 The dataset is a snapshot of customers rather than a full, dated history of customer events.

3 The observed associations do not prove causation.

4 Model performance may change when applied to new customers or different business environments.

5 Churn predictions are estimates and should not be treated as certain outcomes.

14.1 Temporal Analysis

A time-based validation strategy cannot be implemented properly with this dataset alone because it has no dated customer history. The project therefore uses stratified train–test splitting and cross-validation. A production-ready model should be evaluated on historical records with a time-based holdout that reflects the intended prediction date.

15. Future Scope

- Tune hyperparameters and test additional algorithms such as gradient boosting.
- Adjust the decision threshold to balance missed churners against false alarms according to business cost.
- Add dated customer histories to allow time-based validation and churn forecasting.
- Deploy the saved model behind a simple dashboard or application for retention teams.

16. Conclusion

This project investigated telecom customer churn using exploratory analysis, statistical testing, feature engineering and machine learning. Both models clearly outperformed the baseline, which could not identify any churners. Logistic Regression gave the best overall balance (F1-score 0.613, ROC-AUC 0.842, recall 0.794), while Random Forest gave the highest accuracy and precision. Short tenure, month-to-month contracts, higher charges, electronic-check payment and a lack of security or support services were the strongest indicators of churn. The findings can help identify customer groups for further investigation and inform retention experiments.

References

- IBM, Telco Customer Churn dataset: https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv
- scikit-learn documentation: https://scikit-learn.org
- pandas documentation: https://pandas.pydata.org
- SciPy documentation: https://scipy.org